

SAR Automatic Target Recognition Based on Euclidean Distance Restricted Autoencoder

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Abstract—Deep learning algorithms have been introduced into target recognition of synthetic aperture radar (SAR) images for extracting deep features because of its accuracy on various recognition problems with sufficient training samples. However, applying deep structures in recognizing SAR images may suffer lack of training samples. Therefore, a deep learning method is proposed in this study based on a multilayer autoencoder (AE) combined with a supervised constraint. We bind the original AE algorithm with a restriction based on Euclidean distance to use the limited training images well. Moreover, a dropout step is added to our algorithm, which is designed to prevent overfitting caused by supervised learning. Experimental results on the MSTAR dataset demonstrate the effectiveness of the proposed method on real SAR images.

Index Terms—Autoencoder (AE), deep learning, dropout, Euclidean distance restriction, synthetic aperture radar (SAR) imagery, target recognition.

I. INTRODUCTION

SYNTHETIC aperture radars (SAR) can obtain high-resolution images and information of target identification in all weather and full-time; they can also acquire data of large areas. An SAR image contains both shape and scattering information, thereby offering a good representation of objects. Therefore, SAR images are useful tools in target recognition [1]–[3], and automatic target recognition (ATR) of SAR images is significant in both commercial and military areas.

A great many of methods, which can be generally divided into two kinds, have been applied to ATR of SAR images. One is devoted to represent the objects in another subspace in order to obtain discriminative features. For example, principal component analysis (PCA) [4], sparse representation (SR) [2], [5], [36], graphical models [40], and scattering center models [37] are such kind of methods. The other aims to achieve exceptional results using different classification rules, such as

template matching [6], [7], [38], particle swarm optimization (PSO) with Hausdorff distance (HD) [8], and Adaboost classifier [39]. These methods have gained good accuracies on SAR image recognition. However, these traditional algorithms neglect the human learning system, which can learn hierarchical representations of the objects with discriminative information.

Deep learning [9], which is identical to the multilayer physical structure of the human learning system, has been introduced into various areas because of its ability to learn hierarchical features. Deep learning algorithms have achieved state-of-the-art results in numerous problems [10] concerning large-scale data such as classification [11] and inference [12]. For example, high accuracies have been gained from various datasets, such as the MNIST digit dataset, which has 60 000 training samples, and the CIFAR-10 image dataset containing 50 000 training images. Restricted Boltzmann machine (RBM) and autoencoder (AE) are two common unsupervised structures used in deep learning algorithms [13], [14].

Deep learning algorithms have been introduced into problems of remote sensing images, especially hyperspectral images [15], [41], [42]. ATR of SAR images has also employed these deep structures [16], [33], [34], of which [16] and [33] are based on RBM, and [34] employs the AE structure. However, SAR images for learning deep structures are insufficient. In the three-target MSTAR dataset, which is widely used in SAR ATR, only 698 training samples are commonly used as the training data, which is significantly less than those in the MNIST and CIFAR-10 datasets. Thus, the deep learning algorithms cannot perform well with the insufficient training samples in the MSTAR dataset. As a solution, Ma *et al.* [41] attempted to adopt a representation-based classifier to deal with the small quantity of training samples. Zhong [42] controlled the number of convolutional layers and feature maps, and used a global average pooling layer to reduce the number of parameters. Meanwhile, training sample augmentation, a data preprocessing method, was used for SAR image recognition based on deep learning in [16]. Ni [16] augmented training images by ten times with certain rules, such as changing the gray level and scaling.

Data augmentation has the advantages of bringing about data diversity and robust features. However, it also causes other problems. Data augmentation is time consuming in the training stage because of its large amount of training images. Moreover, finding the optimal augmentation method for a given dataset is difficult in data augmentation. In other words, the size of the training dataset for augmentation and which manner of augmentation yields the highest accuracies are unknown. We can just

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try validation experiments with different augmentation methods to select the best one. Therefore, we expect to improve the algorithm itself without such man-made rules.

AE is a common unsupervised structure used in deep learning algorithms [13]. AE is widely used in different areas because it can be learned and extended with optimization structures easily [14]. Some published papers [25], [31] have modified the basic deep learning structure by adding regularization terms to solve certain problems. Rifai [25] added a contractive term to carve a representation that better captures the local directions of variation dictated by the data. Lee [31] encouraged the sparsity of the hidden units by adding a sparse regularization term in the original RBM. In this study, we bind the original AE algorithm with a restriction based on Euclidean distance which introduces supervised information, to make use of the limited training images well. Moreover, a dropout step is added to our algorithm, which is designed to prevent overfitting caused by supervised learning with few training data. Finally, we use a classifier to categorize the features extracted by the Euclidean distance restricted AE.

The rest of the paper is organized into four sections. Section II describes deep learning, AE and stacked AE (SAE) models used in this paper. In Section III, our modified AE structure is proposed. Experimental results in comparison with other methods of SAR ATR are shown in Section IV, and Section V concludes this paper.

II. BRIEF OVERVIEW OF DEEP LEARNING AND AE

A. Deep Learning

Concepts of deep learning are derived from the study of artificial neural networks with multiple hidden layers as early as 1989 [17]. However, due to the lack of proper training algorithms in the early years, people could not harness this powerful model until Hinton proposed his deep learning idea in 2006 [18]. Hinton and Salakhutdinov [18] proposed an unsupervised greedy training algorithm in 2006, which aimed at resolving the optimization problems of deep network structure, leading to a widespread concern in machine learning-related research communities [17], [18].

Deep learning involves a series of models that try to extract deep features of the input data hierarchically with deep neural networks. Among the deep structures, the commonly used AE and RBM networks are layerwise learned by unsupervised training and may be followed by finetuning in a supervised manner. In deep models, high-level features are learned from low-level ones, which are suitable for classification using the hierarchical structures. According to recent papers [19], [20], deep structures can potentially learn more abstract and complex features at higher layers and can provide better approximation to nonlinear functions than shallow models.

Typical deep neural network architectures include belief networks [21], RBM [22], [23], AE [24], [25], predictive sparse decomposition [26], etc. Although available AE variants exist, e.g., regularized AE [25], contractive AE [25], and denoising AE [32], we only adopt the original AE for our model to examine the effectiveness of the Euclidean distance restriction in

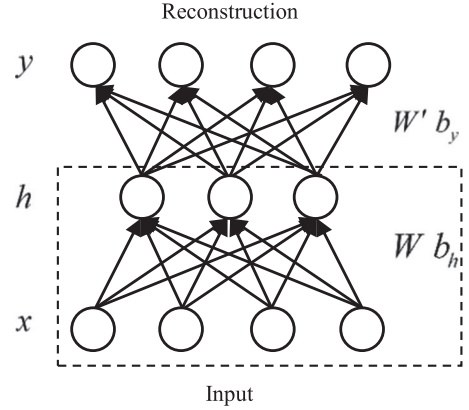


Fig. 1. Structure of a single layer AE. The model learns a hidden feature “ h ” from input “ x ” by reconstructing it on “ y .” The corresponding parameters are denoted in the network.

classifying SAR images; we also choose SAE as the corresponding deep architecture.

B. AE and SAE

In the simplest form, an AE is composed of two parts, an encoder and a decoder. AE was introduced in the 1980s [27], [28] and has been widely used in classification problems [14].

During training, an AE first maps the input to hidden layer and produces the hidden representation $h \in R^{d_h}$. The network corresponding to this step is shown in the boxed part of Fig. 1 and is called an encoder. It has the form

$$h = f(x) = s_f(Wx + b_h) \quad (1)$$

where s_f is a nonlinear activation function, typically a logistic function as $\text{sigmoid}(z) = \frac{1}{1+e^{-z}}$. Other activation functions can also be used, such as the rectified linear unit (ReLU) function [35]. ReLU can be substituted for sigmoid as the activation function for remarkably deep structures because gradient vanishing is a problem in deep neural networks with the sigmoid activation function. The encoder is parameterized by a $d_h \times d_x$ weight matrix W , and a bias vector $b_h \in R^{d_h}$.

Subsequently, reconstruction is done by a decoder by mapping the hidden representation h to an output layer. The reconstruction form is

$$y = g(h) = s_g(W'h + b_y) \quad (2)$$

where s_g is the decoder's activation function, typically either the identity function (yielding linear reconstruction) or a sigmoid function. The parameters in the decoder are a bias vector $b_y \in R^{d_x}$ and a weight matrix W' . We only use the tied weights in our study, in which $W' = W^T$, where T denotes the transpose operation. Thus, we have three parameters to learn: W , b_h , and b_y .

The encoder's output represents the hidden layer feature and the decoder is tuned to reconstruct the initial input x from the encoder's representation $f(x)$ by minimizing the following cost function:

$$J_{AE}(\theta) = \sum_{x \in D_n} L(x, y) = \sum_{x \in D_n} L(x, g(f(x))) \quad (3)$$

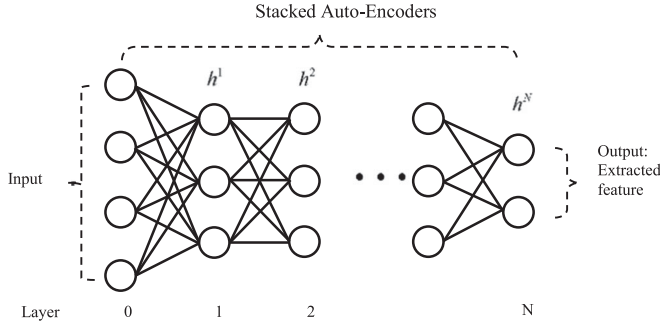


Fig. 2. Typical structure of SAE. Several single AEs are stacked layer by layer to extract deep features.

where L is the reconstruction error function, which is typically the squared error function $L(x, y) = \|x - y\|_2^2$. $\theta = \{W, b_h, b_y\}$ are the parameters in AE and D_n is the training set. The training of AE aims at finding parameters θ that minimize the reconstruction error between the input and reconstruction on training set D_n . Thus, the parameters can be updated by the following gradient descent algorithms:

$$W = W - \eta \frac{\partial J_{AE}(\theta)}{\partial W} \quad (4)$$

$$b_h = b_h - \eta \frac{\partial J_{AE}(\theta)}{\partial b_h} \quad (5)$$

$$b_y = b_y - \eta \frac{\partial J_{AE}(\theta)}{\partial b_y}. \quad (6)$$

SAE has a hierarchical structure constructed by multilayer AEs. Fig. 2 shows a typical SAE structure. The first AE maps inputs in Layer 0 to the first hidden feature in Layer 1. After the first layer is trained, subsequent AE layers are trained via the output of its previous layer. For example, after gaining the output h^1 of Layer 1, we then use h^1 as the input of Layer 2 which maps h^1 to an output h^2 through the reconstruction cost function $L(x, y) = \|x - y\|_2^2$. The entire SAE structure is trained layer by layer, finally obtaining deep feature h^N at the last layer.

III. EUCLIDEAN DISTANCE RESTRICTED AE

Deep learning is widely used because of its ability to extract deep features; however, lacking training samples is a problem when applying deep methods directly to the recognition of SAR images. Augmenting training samples with some rules, such as changing the gray level and rescaling [16], can partially solve the problem; however, dealing with large amount of training images is time-consuming, and the optimal extension rule is hard to determine for a given dataset. Therefore, we expect to improve the algorithm itself for classification with small training set without such man-made rules. Herein, we proposed an advanced algorithm that binds the original AE with a restriction of Euclidean distance, followed by a dropout step to deal with overfitting in supervised learning.

Fig. 3 shows the manner in which our proposed method works both in the training and testing stages in comparison with the original AE.

A. Euclidean Distance Restriction

Assume X represents the training data and x_i ($i = 1, \dots, N$) is the i th training sample, where $x_i \in R^n$ is a column vector and N represents the total number of the training data. The original AE maps input X to the hidden layer H and h_i is the hidden layer of the i th training sample, where $h_i \in R^m$ is also a column vector. Thus, the original AE minimizes the reconstruction error:

$$\min J(\theta) = \min L(x, y) = \sum_{i=1}^N \|x_i - g(f(x_i))\|_2^2 \quad (7)$$

where $h_i = f(x_i) = s_f(Wx_i + b_h)$ represents the encoder and $y_i = g(h_i) = s_g(W^T h_i + b_y)$ is the decoder reconstructing the input from the hidden layer. $\theta = \{W, b_h, b_y\}$ are parameters of the structure.

In our Euclidean distance restricted AE, the hidden layer features from the same target are expected to be more similar than those from different classes. Thus, another term is added—Euclidean distance restriction—to the reconstruction error:

$$\begin{aligned} \min J(\theta) = \min L(x, y) = & \sum_{i=1}^N \|x_i - g(f(x_i))\|_2^2 \\ & + \lambda \cdot \sum_{i=1}^N \sum_{j=1}^N (\|h_i - h_j\|_2 - S_{ij})^2 \end{aligned} \quad (8)$$

where S_{ij} restricts the Euclidean distance of the hidden layer features. When training samples x_i and x_j are from the same target, $S_{ij} = 0$; otherwise $S_{ij} = \alpha$, where α is a constant greater than zero. Through this penalty term, hidden layer features from the same target are more similar due to their shorter distances in feature space. Meanwhile, a large distance is encouraged by S_{ij} when two samples are from different targets.

The parameters of the Euclidean distance restricted AE are updated by the gradient descent algorithm, similar to a normal AE:

$$W = W - \eta \frac{\partial J(\theta)}{\partial W} \quad (9)$$

$$b_h = b_h - \eta \frac{\partial J(\theta)}{\partial b_h} \quad (10)$$

$$b_y = b_y - \eta \frac{\partial J(\theta)}{\partial b_y}. \quad (11)$$

A hierarchical structure is constructed by stacking Euclidean distance restricted AEs layer by layer, which is used for extracting deep features and can be easily learned by the backpropagation (BP) algorithm [17].

Our proposed Euclidean distance restriction can also adjust to a problem in SAR imagery, known as target aspect sensitivity. In passive vision systems, images are mainly formed via diffuse reflection from a noncoherent source of light. Conversely, SAR images are formed from specular reflections of a coherent source, and their characteristics vary quickly with small changes in pose. Thus, target aspect variations lead to scatterers' motion through range cells, which makes the targets in the SAR images quite different when their aspect angles differ. The Euclidean

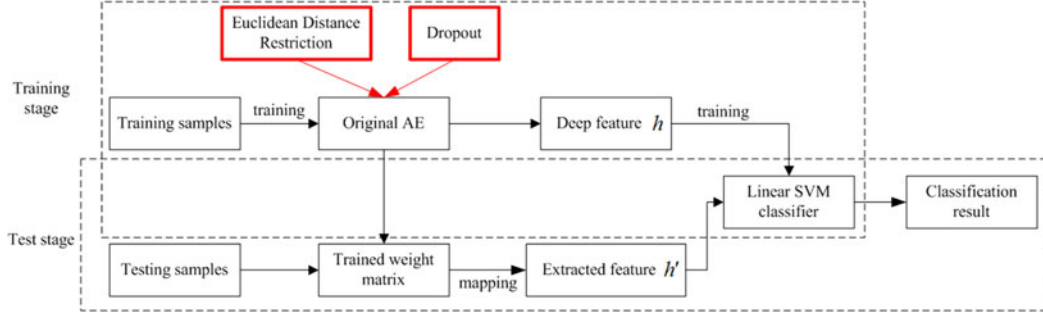


Fig. 3. Structure of our proposed method. Red parts with bold lines represent our improvement of the original AE.

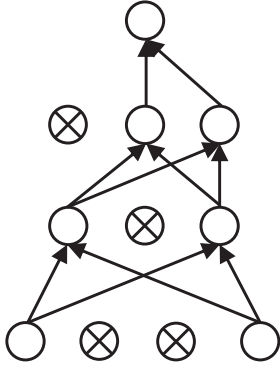


Fig. 4. Example of a dropout neural network model. Crossed units have been dropped.

distance restriction encourages the similarity of extracted features from the same target at different aspect angles, and is useful in ATR of SAR images.

B. Dropout

The recently developed “dropout” is a technique that prevents overfitting and provides a manner of approximately combining various neural network architectures efficiently [29]. Dropout refers to dropping units (hidden and visible) randomly in a neural network, that is, temporarily removing it from the network along with all its incoming and outgoing connections, as shown in Fig. 4. Neurons that are “dropped out” in this manner do not participate in the forward pass and do not contribute in the BP step. This technique reduces the complex coadaptation of neurons, because a neuron cannot rely on the presence of other particular neurons [30]. Thus, overfitting is efficiently prevented through a dropout step.

Consider a deep network with L hidden layers. Let $l \in \{1, \dots, L\}$ index the hidden layers of the network. The non-linear part of the structure departs into two parts, that is, inputs into a layer and outputs from the layer. Let Z^l denote the vector of inputs into layer l , and H^l denotes the vector of outputs from layer l . W^l and b^l are the weights and biases at layer l , respectively. Fig. 5(a) shows that the feed-forward operation of a standard neural network can be described as

$$z_i^{l+1} = W_i^{l+1} H^l + b_i^{l+1} \quad (12)$$

$$h_i^{l+1} = f(z_i^{l+1}) \quad (13)$$

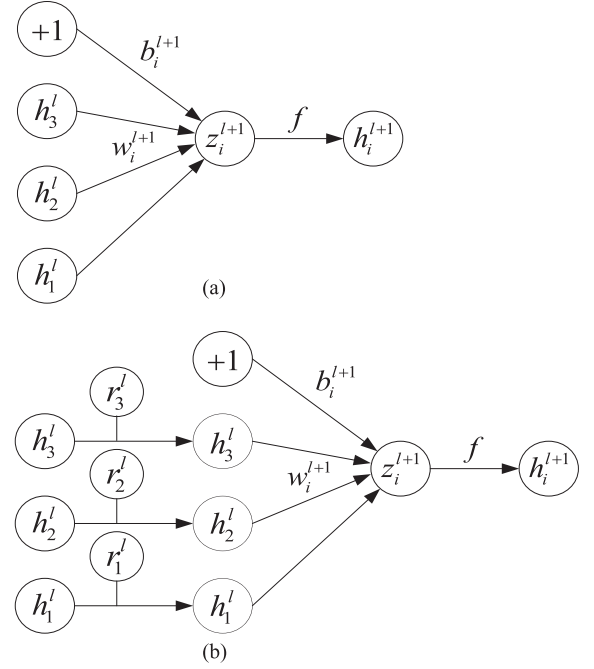


Fig. 5. Comparison of the basic operations of a standard and dropout network.

where $l \in \{1, \dots, L-1\}$, h_i denotes the i th hidden unit and W_i denotes the i th row of the weight matrix W . f is an activation function, such as the sigmoid function used in our algorithm.

As shown in Fig. 5(b), with dropout, the feed-forward operation becomes

$$r_j^l \sim \text{Bernoulli}(p) \quad (14)$$

$$\tilde{h}_j^l = r_j^l \cdot h_j^l \quad (15)$$

$$z_i^{l+1} = W_i^{l+1} \tilde{H}^l + b_i^{l+1} \quad (16)$$

$$h_i^{l+1} = f(z_i^{l+1}). \quad (17)$$

For any layer, r_j^l is an independent Bernoulli random variable which has probability p of being 1. Each r_j^l is sampled and multiplied with the output of that layer h_j^l and then creates the thinned output vector $\tilde{H}^l$. The thinned outputs are used as the input to the next layer. This process is applied at each layer and is learned easily by the BP algorithm.

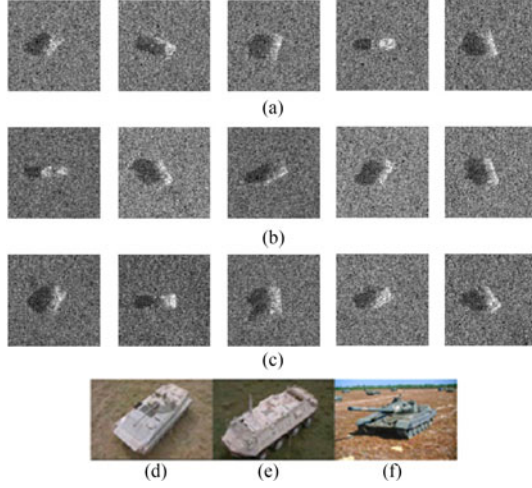


Fig. 6. SAR and optical images of the three-target data to be identified. (a) gives some SAR imagery examples of BMP2 and (d) is its optical image. (b) gives some imagery examples of BTR70 and (e) is its optical image. (c) gives some imagery examples of T72 and (f) is its optical image.

TABLE I
TYPE AND NUMBER OF TRAINING AND TEST SAMPLES
FOR THREE-TARGET DATA

Dataset	BMP2			BTR70		T72	
	C21	9566	9563	C71	132	S7	812
Training samples (17°)	233	0	0	233	232	0	0
Test samples (15°)	196	196	195	196	196	191	195

IV. EXPERIMENTAL RESULTS

A. Data Description

In this study, MSTAR dataset with both three-target data and ten-target data are tested using our proposed Euclidean distance restricted AE. The three-target data contains two types of tanks, T72 and BMP2, and an armored vehicle BTR70. Some SAR and optical imagery examples of these three types of targets are shown in Fig. 6. Each of the targets has full aspect coverage from 0° to 360° and different views at 15° and 17° depression angles. The data at depression 17° are used for training and those at depression 15° for test.

Some details of the training and test samples of the three-target data are listed in Table I. As shown in Table I, there are three different serial numbers for BMP2 and T2. Same as the recognition experiments in [16], we only use the images of BMP2-9563, BTR70-C71, and T72-132 at depression angle 17° as the training data, while all images of BMP2-C21, BMP2-9566, BMP2-9563, BTR70-C71, T72-132, T72-S7, and T72-812 at depression angle 15° are used as the test data.

Besides the three-target data, the ten-target dataset contains the data from another seven ground vehicles. The optical imagery examples of these ten types of targets are shown in Fig. 7. Similar to the recognition experiments on three-target data, we use the images at depression angle 17° without variants as the training data, and all images at depression angle 15° are used

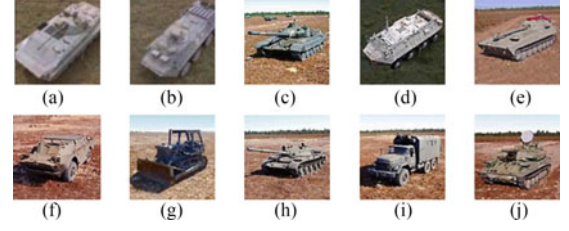


Fig. 7. Optical images of the ten-target data. (a) BMP2 (b) BTR70 (c) T72 (d) BTR60 (e) 2S1 (f) BRDM2 (g) D7 (h) T62 (i) ZIL131 (j) ZSU234

as the test data in the ten-target experiments. Some details of the training and test samples of the ten-target data are listed in Table II.

The amplitudes of all images differ a lot in the MSTAR dataset. We adopt the normalization preprocessing method because huge difference of the amplitudes between training samples may conceal their own differences between the targets using deep learning algorithms to extract features. Each image in both training and test sets is normalized and reshaped to a column vector before training and test in our proposed structure. Except for normalization, we do not use any other preprocessing methods, such as data augmentation, filtering and image segmentation [16], in our experiments.

B. Three-Target Experiments

a) *Experiment Setup*: In the three-target experiments, we used a two-layer Euclidean distance restricted AE, which has 1000 hidden units in the first layer and 400 in the second. The restriction distance is set to $S_{ij} = 10$ when training sample x_i and x_j are from different targets, and the restriction weight is set to $\lambda = 14$. The dropout fraction is set to $\rho = 0.5$ in the dropout step. After extracting deep features, a linear SVM is employed for classification. The linear SVM is chosen as the classifier because there are no kernel parameters to be tuned in the linear SVM, and a good result with a linear classifier demonstrates the linear separability of the features extracted by our proposed method.

b) *Analysis of Our Proposed Method*: In this section, we discuss the effectiveness of our proposed method. We first project the features learned by the original AE and the Euclidean distance restricted AE to the two-dimensional PCA subspace to compare the features qualitatively. As shown in Fig. 8, the features learned by the proposed method are obviously more discriminative than those by the original AE.

In addition, the ratio of between-class distance to within-class distance (BWR) [44] is adopted to measure the features' linear separability learned by our proposed method quantitatively. The BWR is defined as follows:

$$\text{BWR} = L_{\text{between}} / L_{\text{within}} \quad (18)$$

$$L_{\text{within}} = \sum_{q=1}^Q \frac{1}{n_q} \sum_k^{n_q} (x_k^{(q)} - \bar{x}^{(q)})^T (x_k^{(q)} - \bar{x}^{(q)}) \quad (19)$$

TABLE II
TYPE AND NUMBER OF TRAINING AND TEST SAMPLES FOR TEN-TARGET DATA

Dataset	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
Training samples (17°)	233 (C21)	233	232 (132)	255	299	298	299	299	299	299
Test samples (15°)	587 (C21, 9566, 9563)	196	582 (132, S7, 812)	195	274	274	274	273	274	274

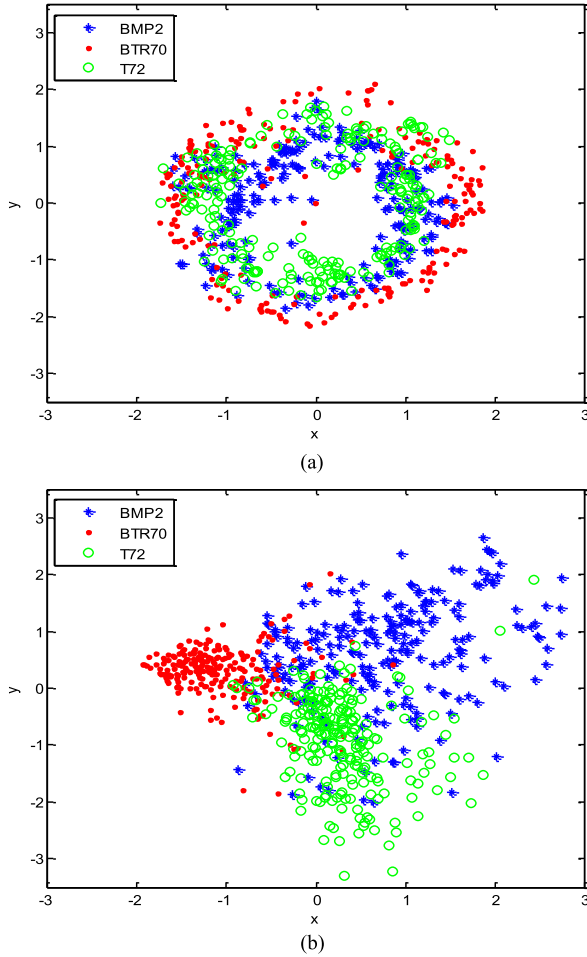


Fig. 8. Two-dimension PCA projection of MSTAR training samples. (a) is learned by original AE, while (b) is learned by Euclidean distance restricted AE.

$$L_{\text{between}} = \sum_{q=1}^Q (\bar{x}^{(q)} - \bar{x})^T (\bar{x}^{(q)} - \bar{x}) \quad (20)$$

where BWR is the BWR value of feature $\{x_k^{(q)}\}_{q=1, k=1}^{Q, n_q}$, L_{between} represents the between-class distance in the feature space, and L_{within} represents the within-class distance in the feature space. Q denotes the number of classes, which is 3 on the three-target MSTAR dataset in our experiment. n_q denotes the number of samples from the q th class and $x_k^{(q)}$ denotes the k th sample from the q th class. $\bar{x}^{(q)}$ is the mean value of all samples from the q th class and $\bar{x}$ represents the mean value of samples from all the classes. The smaller L_{within} and the larger L_{between} are in the feature space, the larger the BWR value of

TABLE III
BWR VALUES OF ORIGINAL IMAGES AND EXTRACTED FEATURES

	Original training images	Features extracted by the Original AE	Features extracted by our proposed method
BWR	1.03	1.04	2.50

the feature is; that means the feature is more linearly separable. The BWR values of the original training images and features that are extracted from both the original AE and our proposed method are listed in Table III. In Table III, the BWR values of the original training images and features extracted from the original AE are similar, and the value of our proposed method is obviously higher than those two. The result indicates that the smaller within-class distance and larger between-class distance are encouraged in the feature space of our method. Therefore, discriminative features are learned from our proposed method because of the Euclidean distance restriction.

c) *Recognition Results:* In this section, we first compare the recognition performance of some feature extraction methods followed by linear SVMs on the three-target MSTAR dataset mentioned above, including directly using original images, original AE, our proposed method without dropout (original AE + Euclidean distance restriction), and our proposed method (original AE + Euclidean distance restriction + dropout). Other SAR recognition methods, including SR-based methods [5], [36], PCA with kernel SVM [4], PSO with HD [8], Adaboost [39], nonnegative matrix factorization (NMF) [43], and some deep learning methods [16], [33], [34] are also compared with our proposed method. Detailed accuracies of different types of vehicles on the three-target dataset via those four feature extraction methods with linear SVM, Adaboost [39] and NMF [43] are listed in Table IV. In Table IV, “Adaboost” represents applying the Adaboost classifier after pose estimation [39]; “NMF” represents extracting features via nonnegative matrix factorization [43]. As shown in Table IV, for AE-based methods (original AE, proposed method without dropout and proposed method), accuracy is improved because of the Euclidean distance restriction and the dropout step. The Euclidean distance restriction can extract discriminative features, therefore the accuracies of proposed method without dropout and the proposed method in the cases both with and without variants (as shown in the fifth and sixth columns in Table IV) are improved in comparison with those of the original AE. Meanwhile, since the dropout step can prevent overfitting [29], as shown in the sixth Column in Table IV, the accuracy of our proposed method in the case with variants only is considerably improved (over 8%) in comparison with that without dropout. This result shows that a better

TABLE IV
DETAILED ACCURACIES OF DIFFERENT TYPES OF VEHICLES ON THE
THREE-TARGET DATASET VIA SOME SAR RECOGNITION METHODS

Vehicle Method	BMP2	BTR70	T72	Without variants	Variants only	Avg.
Original images + LSVM	80.75%	98.47%	93.30%	94.68%	84.14%	88.64%
Original AE + LSVM	87.56%	94.39%	83.51%	93.14%	82.10%	86.81%
Proposed method without dropout + LSVM	91.65%	96.94%	84.19%	96.57%	83.76%	89.23%
Proposed method + LSVM	94.21%	93.88%	94.16%	97.08%	91.94%	94.14%
Adaboost [39]	94.38%	97.96%	97.94%	/	/	96.12%
NMF [43]	92.50%	100%	96.22%	/	/	95.16%

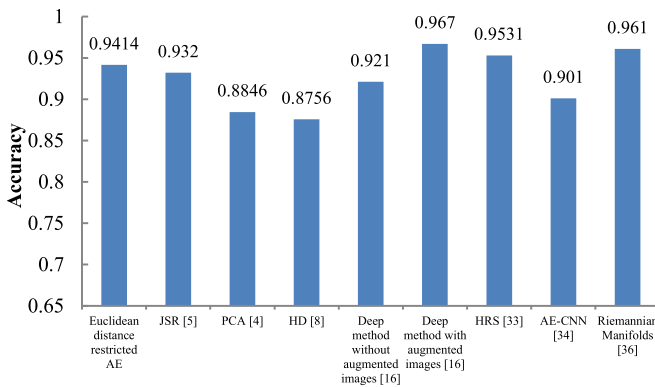


Fig. 9. Three-target accuracies obtained by different SAR ATR methods.

generalization performance is achieved by dropout. The accuracy of Adaboost [39] is higher than ours for its preprocessing methods. Adaboost [39] employs preprocessing methods such as filtering, target segmentation, and pose estimation to obtain more representational features for different targets. NMF [43] uses norm restrictions to encourage sparsity and its accuracy is close to our proposed method.

Correct classification rates of other SAR recognition methods are shown in Fig. 9. In Fig. 9, “Euclidean Distance Restricted AE” represents our proposed method; “JSR” represents joint SR [5]; “PCA” represents PCA with kernel SVM [4]; “HD” represents PSO with HD [8]; “HRS” represents the RBM-based algorithm with several restrictions [33]; “AE-CNN” represents the convolutional neural network (CNN) with sparse AE pre-training [34]; “Riemannian Manifolds” represents SR of monogenic signal via Riemannian manifolds [36]. Results in [16] show the accuracy improvement of data augmentation, which can partially solve the problem of insufficient training images in SAR imagery. However, our proposed method aims to improve the algorithm itself without such man-made rules. In comparison with [16], neither data augmentation nor other preprocessing

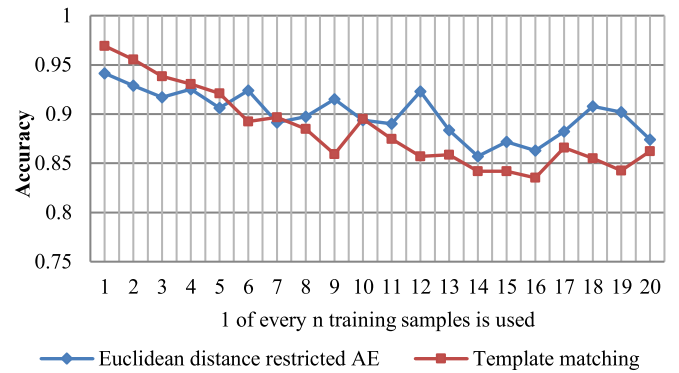


Fig. 10. Accuracies of Euclidean distance restricted AE and template matching. One of every n training samples is used.

methods is used in our proposed method. As shown in Fig. 9, In comparison with some traditional methods without augmented dataset (JSR [5], PCA [4], and HD [8]), our proposed method yields the highest accuracy because of the Euclidean distance restriction and the dropout step. Compared with other deep learning methods without augmented datasets (deep method without augmented images [16], HRS [33], and AE-CNN [34]), the accuracy of our proposed method is higher than those of deep method without augmented images [16] and AE-CNN [34], and is close to that of HRS [33], which also adopts restrictions to obtain representational features. This demonstrates the effectiveness of our restriction. Riemannian manifolds [36] achieves a better performance than our proposed method because of using more information from the complex data, which contains both amplitude and phase information.

In addition, we compare the recognition results of the Euclidean distance restricted AE and the template matching method varying with different numbers of training samples. In this experiment, only one of the training images from every n samples is used with $n = 1, 2, \dots, 20$. Experimental results with different numbers of training samples are shown in Fig. 10. Because of the target aspect sensitivity problem in SAR imagery, the smaller number of training samples leads to the smaller coverage of the aspect angle, which decreases the accuracy considerably in the template matching method. As shown in Fig. 10, the recognition performance of our proposed method is better than that of the template matching method, especially in the cases with small number of training samples. Therefore, our method can adapt to the target aspect sensitivity in SAR imagery to some extent.

d) Analysis on Parameters: In our proposed method, there are three additional parameters compared with the original AE structure, the dropout fraction ρ , the weight of the restriction λ , and the restriction distance α . The parameter ρ represents the probability of each unit being dropped in the network, which varies from 0 to 1 and is typically set to be 0.5 in [29]. Small dropout fraction is not able to prevent overfitting and a large one means too much units being dropped in the structure, which also decreases the performance. Fig. 11 shows the manner in which the dropout fraction ρ affects the classification results. In

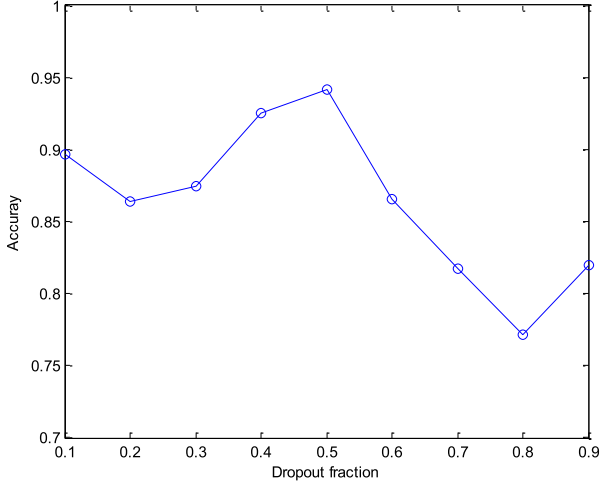


Fig. 11. Accuracies with different dropout fractions.

TABLE V
VALUE OF THE TWO RESTRICTION PARTS

Restriction part	$\sum_{i=1}^N \ x_i - g(f(x_i))\ _2^2$	$\sum_{i=1}^N \sum_{j=1}^N (\ h_i - h_j\ _2 - S_{ij})^2$	Ratio
Value	397.86	24.2206	16.4265

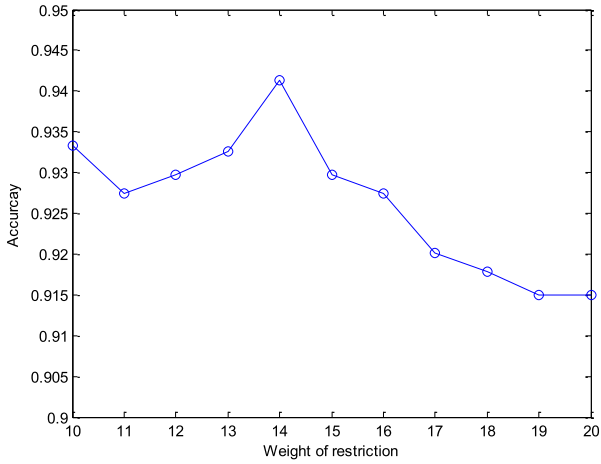


Fig. 12. Accuracies with different restriction weights.

Fig. 11, the results show that the best accuracy is obtained with the parameter $\rho = 0.5$.

In (8), our cost function to be optimized contains two parts, error of reconstruction $\sum_{i=1}^N \|x_i - g(f(x_i))\|_2^2$, and error of distance $\sum_{i=1}^N \sum_{j=1}^N (\|h_i - h_j\|_2 - S_{ij})^2$. We calculate the value of the two parts among the training set. As listed in Table V, the ratio of the two parts is about 16. Fig. 12 shows the variation of the classification accuracy of our method with the value of parameter $\lambda \in [10, 20]$. We can see the optimal value is $\lambda = 14$, and either larger or smaller value of λ decreases the classification accuracy. A too large value of λ represents a large ratio of $\sum_{i=1}^N \sum_{j=1}^N (\|h_i - h_j\|_2 - S_{ij})^2 / \sum_{i=1}^N \|x_i - g(f(x_i))\|_2^2$,

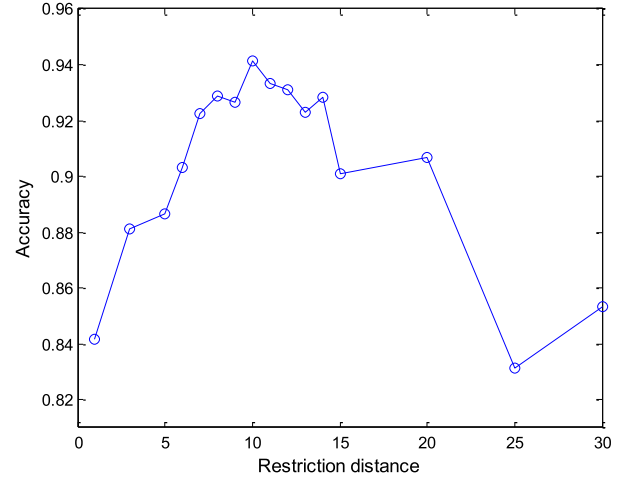


Fig. 13. Accuracies with different restriction distances.

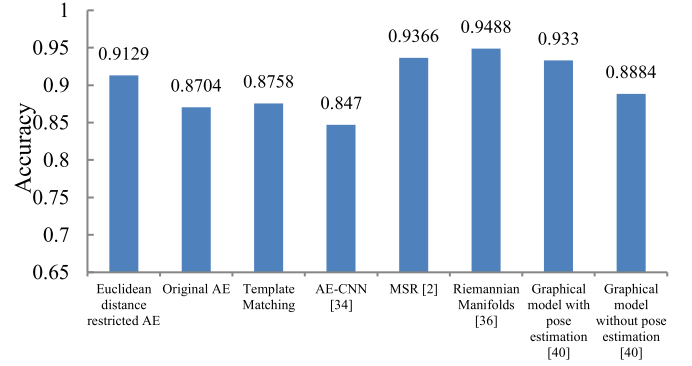


Fig. 14. Average accuracies of different methods on the ten-target dataset.

which is not good for generalization performance, and a too small one is not able to provide enough restriction.

The restriction distance S restricts the hidden layer distance, where $S_{ij} = \alpha$ when training sample x_i and x_j are from different targets, and otherwise $S_{ij} = 0$. The maximum of the hidden layer distance depends on the number of hidden units. It is equivalent to $\sqrt{m}$, where m denotes the number of hidden units. In our proposed method, the size of the structure is set to 1000–400, thus the maximum of the hidden layer distance is $\sqrt{1000} \approx 31.6$. The experimental results of different restriction distances $S \in [1, 30]$ are shown in Fig. 13. We can see that the optimal value is $S = 10$. A too small distance is not able to function well because of its weak restriction, while a too large one may cause overfitting, which also decreases the accuracy.

C. Ten-Target Experiments

a) *Experiment Setup*: In ten-target experiments, we use a two-layer Euclidean distance restricted AE, which has 1200 hidden units in the first layer and 800 in the second. The restriction distance is set to $S_{ij} = 10$ when training sample x_i and x_j are from different targets, and the restriction weight is set to $\lambda = 14$. We set the dropout fraction $\rho = 0.5$ in the dropout step. The softmax is employed for classification after extracting features, because it is hard for the linear SVM to classify ten different classes.

TABLE VI
DETAILED ACCURACIES OF DIFFERENT TYPES OF VEHICLES ON THE TEN-TARGET DATASET VIA SOME SAR RECOGNITION METHODS

Different methods	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234	Avg.
Proposed method	87.90%	92.86%	79.55%	91.79%	93.80%	96.72%	98.91%	94.14%	94.53%	99.64%	91.29%
Original AE	82.45%	90.82%	67.35%	87.18%	91.61%	95.62%	97.08%	94.51%	92.34%	98.91%	87.04%
Template matching	86.37%	92.35%	69.93%	91.79%	85.77%	88.69%	98.18%	96.70%	93.07%	97.45%	87.58%
MSR [2]	88.1%	99.0%	78.2%	97.8%	97.1%	99.3%	96.9%	99.3%	97.4%	99.6%	92.79%
Riemannian Manifolds [36]	88.3%	96.9%	82.1%	98.5%	96.7%	99.3%	99.6%	97.8%	99.6%	99.6%	94.88%
Graphical model with pose estimation [4]	95%	94%	96%	93%	87%	85%	98%	93%	94%	96%	93.30%
Graphical model without pose estimation [4]	89%	91%	93%	92%	77%	88%	89%	88%	88%	91%	88.84%

b) Recognition Results: In this section, we compare our ten-target results with other SAR recognition methods, including template matching, CNN with sparse AE pretraining [34], SR of monogenic signal [2], SR of monogenic signal via Riemannian manifolds [36], and discriminative graphical models [40]. Average accuracies of different methods on ten-target dataset are shown in Fig. 14, in which “Euclidean distance restricted AE” represents our proposed method; “Original AE” represents the AE without restrictions; “AE-CNN” represents CNN with sparse AE pretraining; “MSR” represents SR of monogenic signal; “Riemannian Manifolds” represents SR of monogenic signal via Riemannian manifolds; and “Graphical model” represents discriminative graphical model. Detailed accuracies of different types of vehicles on ten-target dataset are also listed in Table VI. We can see the classification accuracy of our proposed method is higher than the original AE because of the Euclidean distance restriction and the dropout step. In comparison with another deep learning method, AE-CNN [34], our proposed method also yields a better performance. Meanwhile, our accuracy is also higher than those of template matching and graphical model without pose estimation [40], which does not adopt preprocessing methods as we do. The results demonstrate that our proposed method can deal with the small training data in SAR imagery. The accuracies of both MSR [2] and Riemannian manifolds [36] are better than our proposed method because of their usage of complex SAR images, which contains both amplitude and phase information. Graphical model with pose estimation [40] obtains a better performance than our proposed method for using the preprocessing method of pose estimation.

D. Results on Different Sizes

Different sizes of deep learning structures have significant effect on classification results, which is an important problem in applications of deep learning. Herein, we give a detailed analysis on sizes of our proposed method. Fig. 15 shows the one-layer classification results varying with the number of hidden units in our proposed method. There are few differences when the numbers of hidden units are more than 1000. We set our structure with 1000 hidden units in the first layer, because the 1000 hidden units can describe the training data well. Fig. 16 shows the results with different numbers of hidden units in the second layer. We can see the second layer with 400 hidden units is suitable for the data, and a too large number of hidden units leads to the decreasing of the classification accuracy of our proposed method. Classification results with different sizes of our proposed method and the original AE are listed in Table

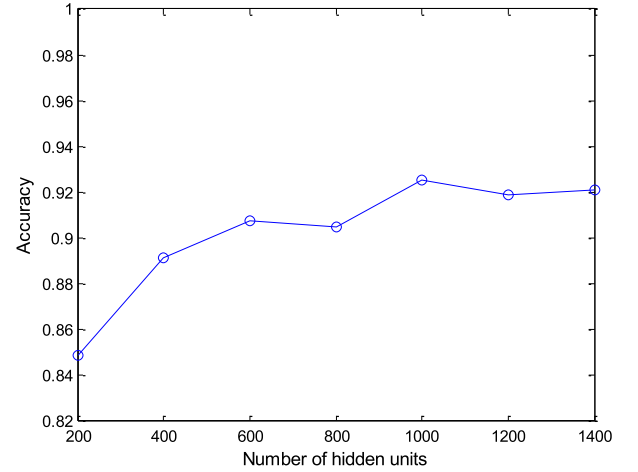


Fig. 15. Accuracies of different numbers of our structure with only one layer.

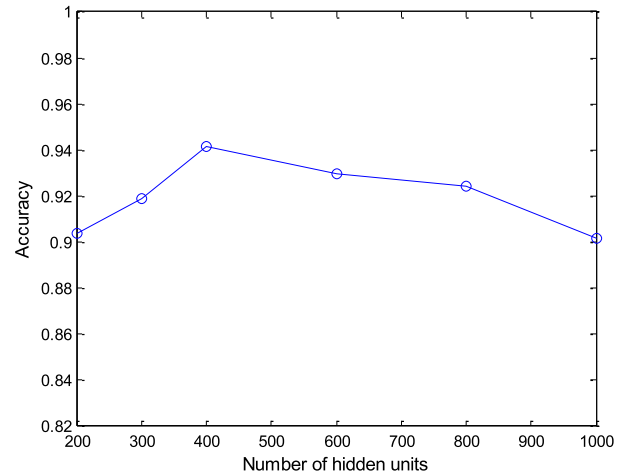


Fig. 16. Accuracies of different numbers of hidden units in the second layer with 1000 first layer hidden units.

VII. In Table VII, “1000–400” means a structure with 1000 hidden units in the first layer and 400 in the second, which is the structure we used in Fig. 10. “1000-400-200” means a structure with 1000 hidden units in the first layer, 400 in the second, and 200 in the third, and “400-400-200” means a structure with 400 hidden units in the first layer, 400 in the second, and 200 in the third. The number of hidden units of the original AE is also denoted in this way in Table VII. Compared with the original AE, our proposed method obtains higher accuracies with both

TABLE VII
ACCURACIES OF DIFFERENT SIZES OF OUR PROPOSED METHOD AND THE ORIGINAL AE ON THREE-TARGET DATASET

Method	Different sizes	Accuracy
Original AE	400-400	86.81%
	400-400-200	85.71%
	300-200-150	86.52%
Our proposed method	1000-400	94.14%
	1000-400-200	83.15%
	400-400-200	88.50%

TABLE VIII
ACCURACIES OF DIFFERENT SIZES OF OUR PROPOSED METHOD AND THE ORIGINAL AE ON TEN-TARGET DATASET

Method	Different sizes	Accuracy
Original AE	800-600	86.29%
	800-600-400	87.04%
	1200-800	91.29%
Our proposed method	800-600-400	91.07%

two-layer and three-layer structures because of the Euclidean distance restriction. The accuracies of the three-layer structures are lower than those of the two-layer structures with both of our proposed method and the original AE, which shows the commonly used 698 training images in the three-target MSTAR dataset is too small for a structure deeper than two layers. As shown in Table VII, the small sizes of the three-layer structures can obtain higher accuracies than the large sizes, because the small sizes cause less overfitting.

In addition, deeper networks are tried on ten-target MSTAR dataset, which contains 2746 training images, larger than the three-target dataset. The experimental results are shown in Table VIII. Compared with the original AE, our proposed method also obtains higher accuracies with both two-layer and three-layer structures because of the Euclidean distance restriction. As shown in Table VIII, we can also see that the deviations between three-layer's results and two-layer's results on the ten-target dataset are smaller than those on the three-target dataset, which indicates that a deeper structure can be used on a larger dataset. As a result, the deeper structures are able to be used if the dataset is large enough.

V CONCLUSION

In this paper, we proposed an advanced deep learning algorithm—Euclidean distance restricted AE—for SAR ATR problems. Specifically, our proposed algorithm combined AE structure with Euclidean distance restriction and dropout to solve the absence of training samples in SAR imagery. The proposed method can extract discriminative features, and can adapt to target aspect sensitivity to some extent. The experimental results show the effect of our Euclidean distance restricted AE on SAR ATR problems compared with other existing methods.

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